THE DERIVED SECTIONS WITH SUPPORT ARE COMPUTED BY THE DERIVED TENSOR PRODUCT WITH THE SECTIONS OF THE STRUCTURE SHEAF, IN FINITE DIMENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-local-cohomology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The description of the sections with support in a specialization-stable subset as the filtered colimit of the sections with support in the closed subsets it contains. Author lemma lemma-adjoint-ext, whose statement is reproduced verbatim as a context span of this card (source lines 996-1004). Combined with the commutativity of directed colimits with direct sums, gives that the functor commutes with direct sums so that the map may be checked on shifts of the ring.
- K-flat complexes as a model for the derived tensor product and the total complex of a double complex. Used at the start of the counted proof (source lines 1180-1184). Provides the explicit representative of the derived tensor product on which the canonical map is built.
- The reduction of an isomorphism of functors commuting with direct sums to the shifts of the ring. Invoked inside the counted proof at source lines 1196-1198 as a cross-chapter pointer. Reduces the verification of the displayed identity to the case of a shift of the ring, where it is immediate.

Source reference. Original reference dualizing-lemma-local-cohomology-adjoint: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 335–335. Navigation pointer only: the counted text refers to the Stacks item dualizing-lemma-local-cohomology-adjoint, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:local-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference more-algebra-remark-P-resolution: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1196–1196. Navigation pointer only: the counted text refers to the Stacks item more-algebra-remark-P-resolution, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let A be a Noetherian ring. Let $T \subset \mathrm{Spec}(A)$ be a subset stable under specialization. Assume A has finite dimension. Then*

$$R\Gamma_T(K) = R\Gamma_T(A) \otimes^{\mathbf{L}}_A K$$

for $K \in D(A)$. For $K, L \in D(A)$ we have

$$R\Gamma_T(K \otimes^{\mathbf{L}}_A L) = K \otimes^{\mathbf{L}}_A R\Gamma_T(L) = R\Gamma_T(K) \otimes^{\mathbf{L}}_A L = R\Gamma_T(K) \otimes^{\mathbf{L}}_A R\Gamma_T(L)$$

If K or L is in $D_T(A)$ then so is $K \otimes^{\mathbf{L}}_A L$.

Lemma 0.2. *Let A be a Noetherian ring. Let $T \subset \mathrm{Spec}(A)$ be a subset stable under specialization. For any object K of $D(A)$ we have*

$$H^i(RH^0_T(K)) = \mathrm{colim}_{Z \subset T \text{ closed}} H^i_Z(K)$$

Proof. By construction we may represent $R\Gamma_T(A)$ by a complex $J^\bullet$ in $\mathrm{Mod}_{A,T}$. Thus if we represent K by a K-flat complex $K^\bullet$ then we see that $R\Gamma_T(A) \otimes^{\mathbf{L}}_A K$ is represented by the complex $\mathrm{Tot}(J^\bullet \otimes_A K^\bullet)$ in $\mathrm{Mod}_{A,T}$. Using the map $R\Gamma_T(A) \to A$ we obtain a map $R\Gamma_T(A) \otimes^{\mathbf{L}}_A K \to K$. Thus by the adjointness property of $R\Gamma_T$ we obtain a canonical map

$$R\Gamma_T(A) \otimes^{\mathbf{L}}_A K \longrightarrow R\Gamma_T(K)$$

factoring the just constructed map. Observe that $R\Gamma_T$ commutes with direct sums in $D(A)$ for example by Lemma 0.2, the fact that directed colimits commute with direct sums, and the fact that usual local cohomology commutes with direct sums (for example by Dualizing Complexes, Lemma dualizing-lemma-local-cohomology-adjoint). Thus by More on Algebra, Remark more-algebra-remark-P-resolution it suffices to check the map is an isomorphism for $K = A[k]$ where $k \in \mathbf{Z}$. This is clear.

The final statements follow from the result we've just shown by transitivity of derived tensor products. □